ChiliDoc

AI-Based Chili Leaf Disease Detection and Treatment Recommendation System

IT41033 - Nature Inspired Algorithms

Final Project Report

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Abstract

Plant diseases are a biggest problem in agriculture field because that can reduce yield and lead to financial damages. **Chili** (*Capsicum annuum*) is an important crop that can be affected by bacterial, fungal and nutritional problems. Identifying these diseases by manually can take a lot of time and may cause incorrect decisions.

This project presents **ChiliDoc** is a deep learning system that compare two approaches for detecting chili plant diseases, a custom **Convolutional Neural Network (CNN)** and a **Vision Transformer (ViT)**. The dataset include **1852 images** from six categories such as Bacterial Spot, Cercospora Leaf Spot, Curl Virus, Healthy Leaf, Nutrition Deficiency and White Spot.

The **CNN** model was trained for 20 epochs and that achieved a validation accuracy of **69.69%.** The **ViT** was trained for 5 epochs and that achieved a high evaluation accuracy **of 99.38%.** The results show that although **CNNs** are lighter and require fewer computational resources, the **ViT** provides better performance by using self-attention to understand important image features. This makes **ViT** a strong option for real time chili disease detection.

1. Introduction

Chili is an important crop used in the spice, food and pharmaceutical industries. However chili plants can be affected by environmental problems, insects and different diseases. Traditionally farmers and experts identify diseases by looking at the leaves. This method can be slow, especially for large farms and mistakes can lead to incorrect treatment.

Computer vision and deep learning can help to increase disease detection. **Convolutional Neural Networks (CNN)** are widely used for image classification because they can automatically learn important features from images. **Vision Transformers (ViT)** use self attention to understand different parts of an image and that can capture relationships between image regions.

This project develops and compares a **custom CNN model and a Vision Transformer (ViT) model** as part of the **ChiliDoc** disease detection system. The project evaluates their training performance, validation accuracy, precision and classification results.

2. Literature Review

Computer vision has become an important technology in precision agriculture. Earlier plant disease detection methods mainly used manually selected features such as **SIFT** and **HOG**, together with machine learning algorithms such as **SVM** and **Random Forest**. These methods worked well in controlled environments but had limitations when images contained different lighting conditions, overlapping leaves, and background noise.

2.1 Evolution of Convolutional Neural Networks (CNN) in Plant Pathology

Deep Convolutional Neural Networks (CNN) reduced the need for manual feature extraction. CNNs can automatically learn features such as colors, edges, textures and leaf disease patterns from images. Different studies using datasets such as **PlantVillage** have shown that CNNs can be effectively used for automated plant disease classification.

Recent research has also focused on improving CNN performance while reducing computational requirements. Studies have explored different CNN architectures and multi-scale feature extraction methods to identify small and complex disease patterns. Research has also shown that the choice of optimization method can affect CNN training performance and classification accuracy.

2.2 Shift Toward Attention-Based Models

Although CNNs are effective at learning local image features, they may have difficulties understanding relationships between distant regions of an image. This can be important for plant disease detection because some diseases affect both small areas and the overall structure of a leaf.

To overcome this limitation, recent research has focused on **Vision Transformers (ViTs)** and other attention-based models. ViTs divide an image into smaller patches and process these patches as tokens. The self attention mechanism allows the model to identify relationships between different parts of the image.

Recent studies have shown that transformer-based models can provide strong performance in plant disease and pest detection. They can capture both local disease patterns and wider relationships within the image. Research has also explored combining deep learning with precision agriculture systems to provide faster and more useful disease detection for farmers.

Overall, previous research shows a clear development from traditional machine learning methods to CNN based approaches and more recently, attention based Vision Transformer models. This development provides a strong foundation for comparing **CNN and ViT models for chili leaf disease classification** in the ChiliDoc project.

3. Methodology

3.1 Dataset Profile & Exploratory Distribution

The image corpus comprises a total of **1852 distinct leaf images** partitioned across six localized diagnostic states.

The categorical distribution across classes is organized as follows:

- **Bacterial_Spot:** 156 images
- **Cercospora_Leaf_Spot:** 180 images
- **Curl_Virus:** 423 images
- **Healthy_Leaf:** 458 images
- **Nutrition_Deficiency:** 444 images
- **White_Spot:** 195 images

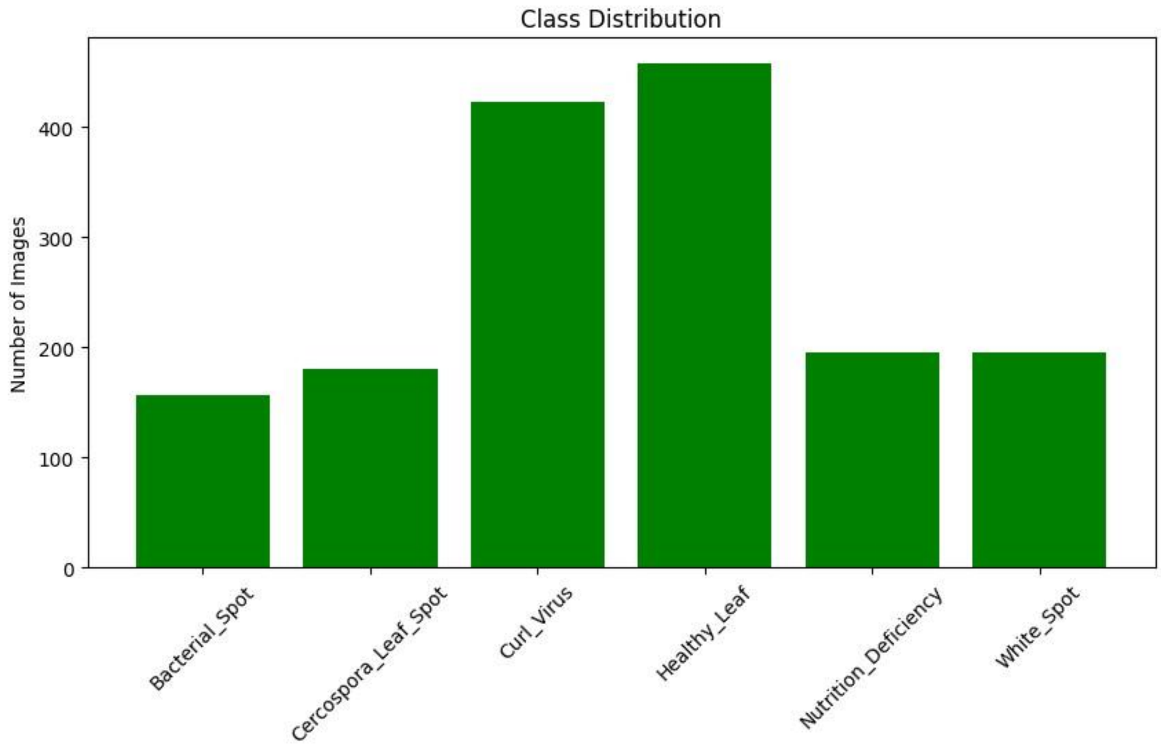


Figure 1: Dataset Exploratory Class Distribution Bar Chart

3.2 Preprocessing and Augmentation Methods

- **CNN**

Images were resized to **224 × 224 pixels** and pixel values were scaled between [0,1].

Data augmentation included **20° rotation, 0.2 zoom and horizontal flipping**.

- **ViT**

Images were resized and converted into tensors.

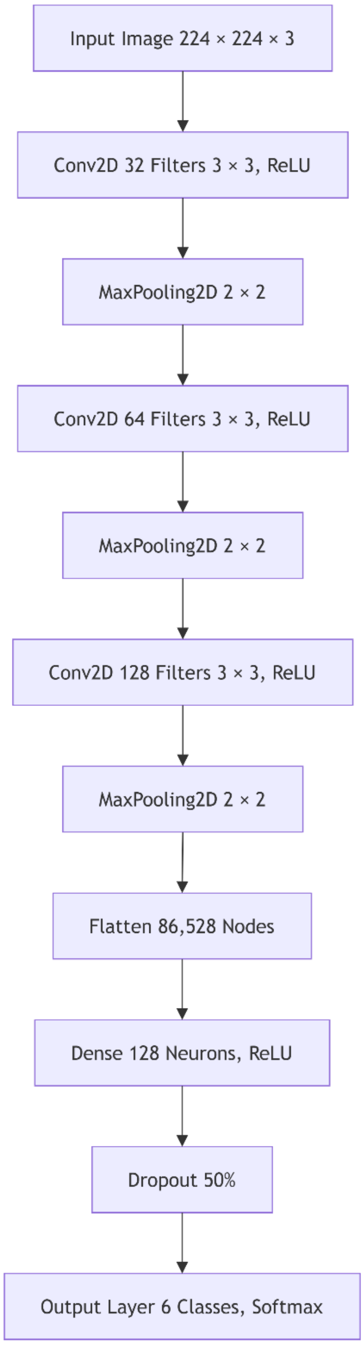
The images were divided into **small patches**, which were then processed by the **Vision Transformer (ViT) algorithm**.

3.3 Structural Network Architectures

3.3.1 Custom CNN Blueprint

The custom CNN contains **three convolutional layers** with **max-pooling layers**. These layers help the model learn important image features and reduce the image size during processing.

Figure 2: Sequential CNN Layout Diagram



3.3.2 Vision Transformer (ViT) Model Structure

The ViT architecture does not use standard convolutional layers. The 224× 224 image is divided into 16 × 16 structural patches. These patches are convert into vector representation and combined with positional embedding. Then the model processes them using **Multi Head Self Attention (MHSA)** to identify important features and relationships within the image.

4. Results

4.1 CNN Model Structure

The parameters used in each layer are detailed below:

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_2 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten (Flatten)	(None, 86528)	0
dense (Dense)	(None, 128)	11,075,712
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 6)	774

Total params: 11,169,734 (42.61 MB)
Trainable params: 11,169,734 (42.61 MB)
Non-trainable params: 0 (0.00 B)

4.2 Model Training History Records

4.2.1 Custom CNN Run Logs (20 Epochs)

```
Epoch 1/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 374s 9s/step - accuracy: 0.4110 - loss: 1.6649 - val_accuracy: 0.3094 - val_loss: 1.4056
Epoch 2/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 38s 932ms/step - accuracy: 0.6154 - loss: 1.0268 - val_accuracy: 0.7625 - val_loss: 0.7357
Epoch 3/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 34s 821ms/step - accuracy: 0.7514 - loss: 0.6680 - val_accuracy: 0.6812 - val_loss: 0.7195
Epoch 4/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 38s 929ms/step - accuracy: 0.7949 - loss: 0.5578 - val_accuracy: 0.6969 - val_loss: 0.7430
Epoch 5/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 34s 835ms/step - accuracy: 0.8516 - loss: 0.4408 - val_accuracy: 0.7344 - val_loss: 0.5647
Epoch 6/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 35s 853ms/step - accuracy: 0.8462 - loss: 0.4526 - val_accuracy: 0.6719 - val_loss: 0.9133
Epoch 7/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 34s 830ms/step - accuracy: 0.8539 - loss: 0.4166 - val_accuracy: 0.7937 - val_loss: 0.4864
Epoch 8/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 36s 873ms/step - accuracy: 0.8749 - loss: 0.3629 - val_accuracy: 0.7406 - val_loss: 0.6895
Epoch 9/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 34s 831ms/step - accuracy: 0.8656 - loss: 0.3964 - val_accuracy: 0.7469 - val_loss: 0.6423
Epoch 10/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 38s 935ms/step - accuracy: 0.9122 - loss: 0.2541 - val_accuracy: 0.8281 - val_loss: 0.4173
Epoch 11/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 34s 827ms/step - accuracy: 0.9215 - loss: 0.2186 - val_accuracy: 0.7844 - val_loss: 0.5361
Epoch 12/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 35s 857ms/step - accuracy: 0.9394 - loss: 0.2099 - val_accuracy: 0.7688 - val_loss: 0.6896
Epoch 13/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 34s 830ms/step - accuracy: 0.9316 - loss: 0.1995 - val_accuracy: 0.6500 - val_loss: 1.4296
Epoch 14/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 34s 842ms/step - accuracy: 0.8850 - loss: 0.3184 - val_accuracy: 0.7812 - val_loss: 0.6229
Epoch 15/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 35s 846ms/step - accuracy: 0.9355 - loss: 0.1890 - val_accuracy: 0.7750 - val_loss: 0.5675
Epoch 16/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 35s 846ms/step - accuracy: 0.9340 - loss: 0.1868 - val_accuracy: 0.7812 - val_loss: 0.6092
Epoch 17/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 36s 867ms/step - accuracy: 0.9534 - loss: 0.1556 - val_accuracy: 0.7688 - val_loss: 0.7468
Epoch 18/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 34s 833ms/step - accuracy: 0.9394 - loss: 0.1532 - val_accuracy: 0.8313 - val_loss: 0.4917
Epoch 19/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 36s 870ms/step - accuracy: 0.9433 - loss: 0.1713 - val_accuracy: 0.8219 - val_loss: 0.5621
Epoch 20/20
41/41 ━━━━━━━━━━━━━━━━━━━━ 35s 849ms/step - accuracy: 0.9363 - loss: 0.1681 - val_accuracy: 0.7594 - val_loss: 0.8821
```

4.2.2 ViT Training Loss Dynamic (5 Epochs)

```
••• Epoch 1/5 - Loss: 0.2944
    Epoch 2/5 - Loss: 0.0271
    Epoch 3/5 - Loss: 0.0193
    Epoch 4/5 - Loss: 0.0079
    Epoch 5/5 - Loss: 0.0086
```

CNN Performance: Reference the dual plots showing Model Accuracy and Model Loss variations across 20 validation checks.

ViT Performance: Reference the training curve mapping the loss reduction from 0.2944 down to 0.0086.

Figure 3: CNN Training Accuracy Curve

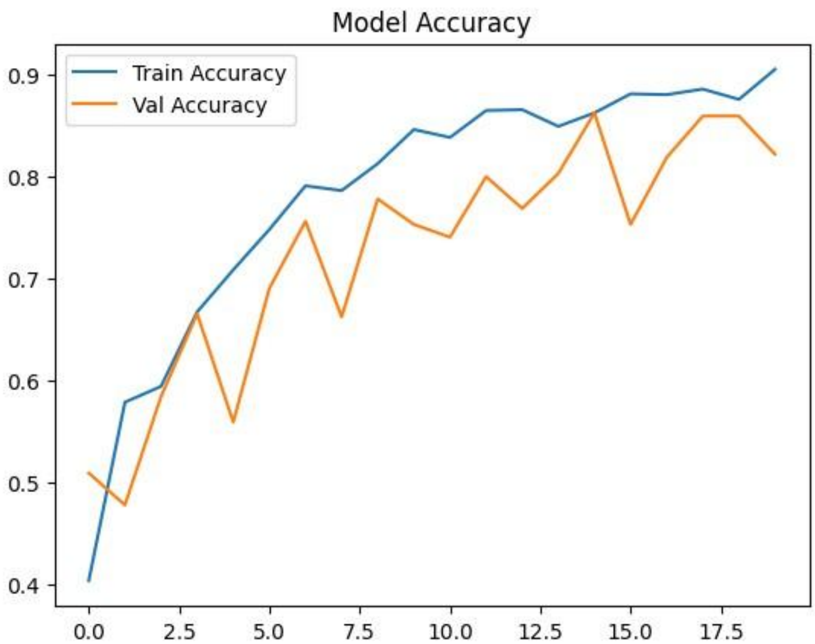
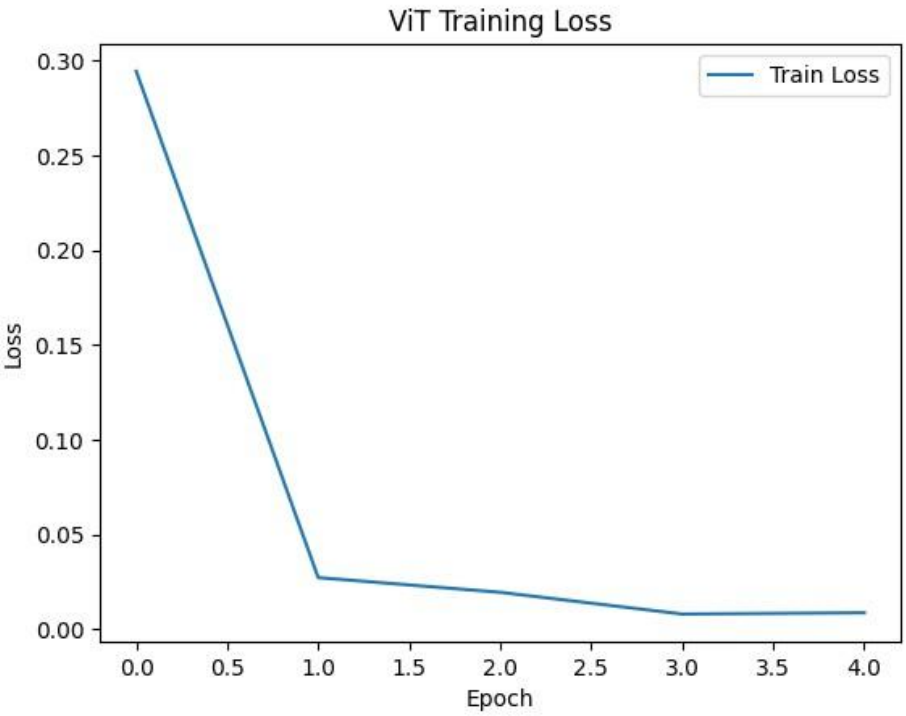


Figure 4: CNN Training Loss Curve

Figure 5: ViT Training Loss Curve



5. Final Evaluation & Model Comparison

5.1 Model Performance Analysis

The performance of the CNN and ViT models was evaluated using the validation dataset. The results are presented below.

Custom CNN Evaluation Metrics (Global Accuracy: 69.69%)

	precision	recall	f1-score	support
Bacterial_Spot	0.14	0.03	0.05	31
Cercospora_Leaf_Spot	0.00	0.00	0.00	36
Curl_Virus	0.26	0.56	0.36	84
Healthy_Leaf	0.23	0.05	0.09	91
Nutrition_Deficiency	0.06	0.03	0.04	39
White_Spot	0.10	0.26	0.15	39
accuracy			0.20	320
macro avg	0.13	0.15	0.11	320
weighted avg	0.17	0.20	0.15	320

Vision Transformer (ViT) Evaluation Metrics (Global Accuracy: 99.38%)

	precision	recall	f1-score	support
Bacterial_Spot	0.97	0.97	0.97	34
Cercospora_Leaf_Spot	0.97	0.97	0.97	37
Curl_Virus	1.00	1.00	1.00	93
Healthy_Leaf	1.00	1.00	1.00	94
Nutrition_Deficiency	1.00	1.00	1.00	30
White_Spot	1.00	1.00	1.00	34
accuracy			0.99	322
macro avg	0.99	0.99	0.99	322
weighted avg	0.99	0.99	0.99	322

5.2 Confusion Matrix Analysis

The confusion matrices show the differences between the actual classes and the classes predicted by each model.

Figure 6: CNN Confusion Matrix

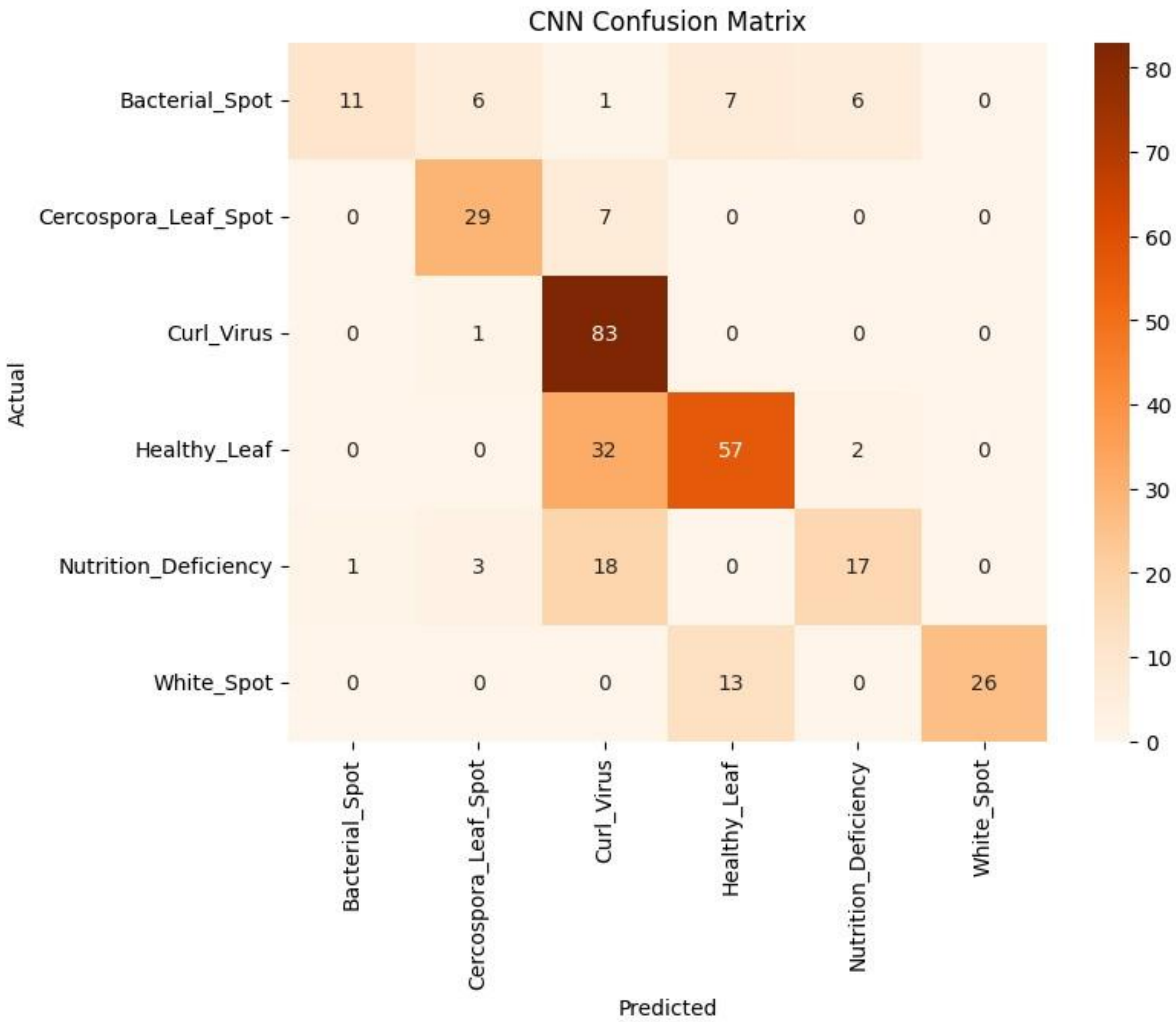
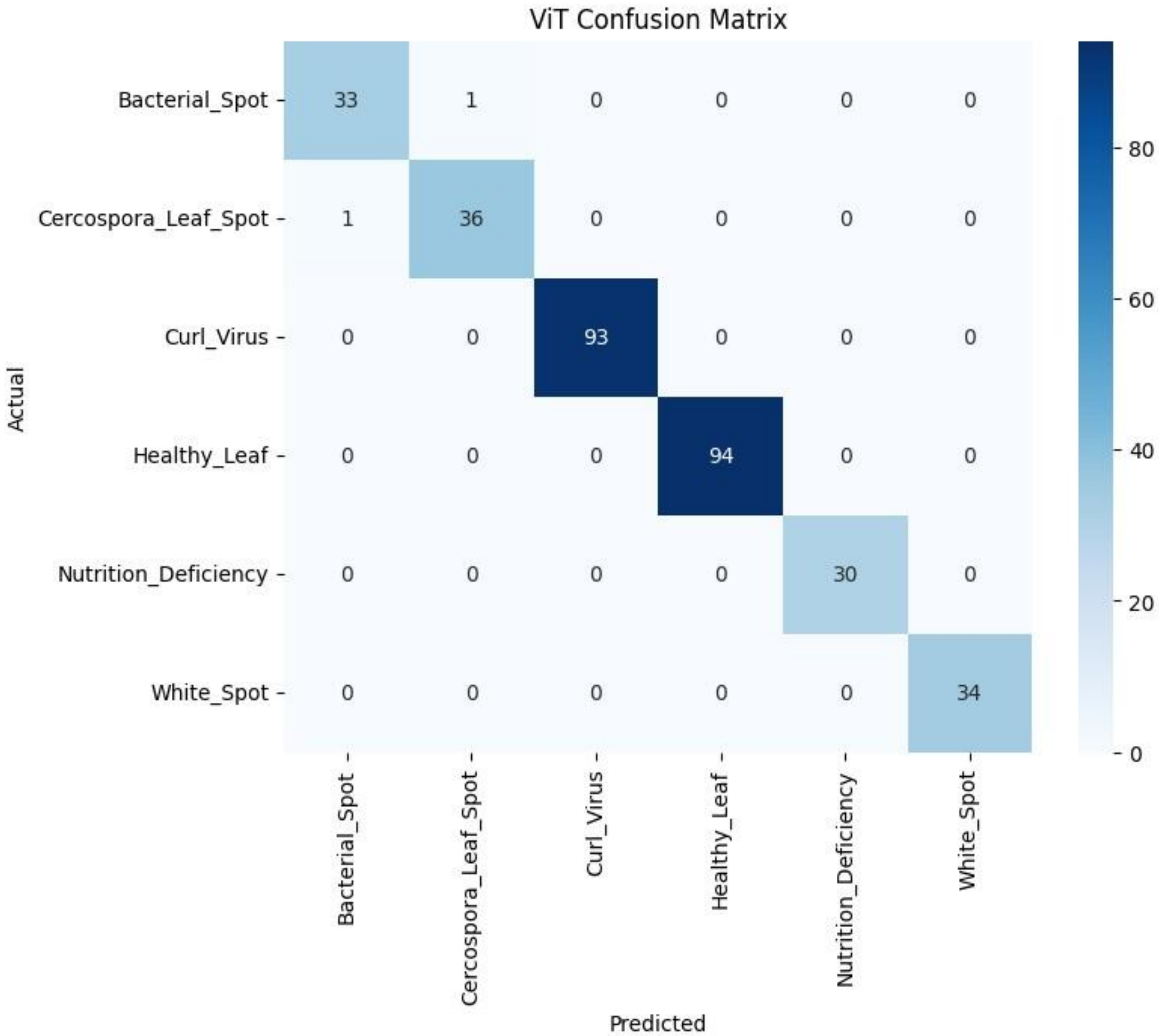


Figure 7: ViT Confusion Matrix

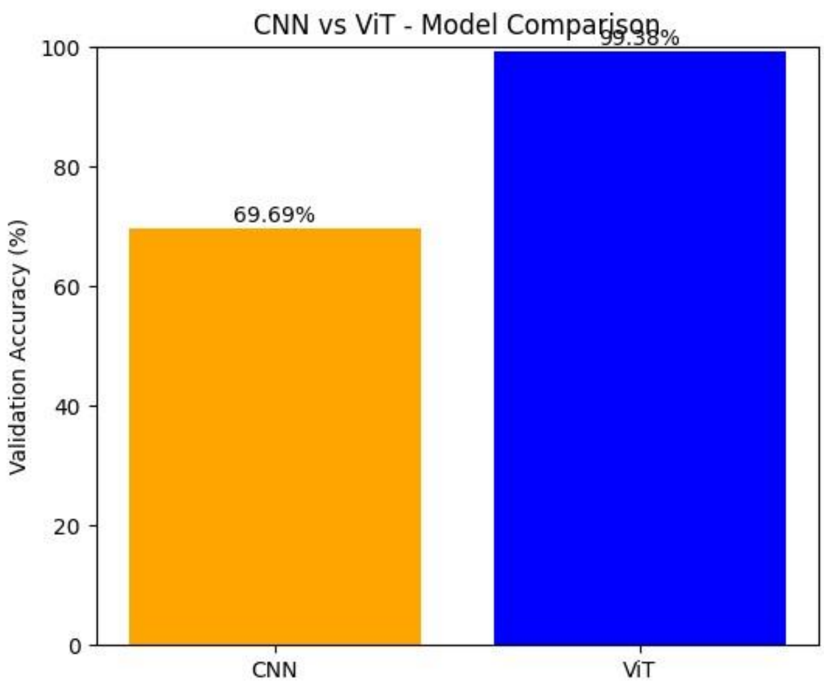


5.3 Model Comparison Summary

The final performance results of the CNN and ViT models are summarized below using the same validation dataset.

Architectural Metric	Architectural Metric	Vision Transformer (ViT)
Final Accuracy Score	69.69%	99.38%
Training Epoch Duration	20 Epochs	5 Epochs
Primary Weakness	Spatial information loss	High computational requirements

Figure 8: Final Evaluation Comparison Bar Chart



6. Discussion

The results show a clear performance difference between the **CNN** and **ViT** models. The custom
CNN achieved 69.69% validation accuracy. The **CNN** had difficulty distinguishing Healthy_Leaf and Nutrition_Deficiency images and often classified them as Curl_Virus. This may be because the **CNN** focuses more on general image features and can lose some small details during pooling.

In comparison, the **Vision Transformer** achieved **99.38%** validation accuracy after only 5 training epochs. The ViT was able to identify small disease patterns in different parts of the leaf by using image patches and self attention. This helped the model understand both local details and the overall leaf structure.

However, the **ViT** requires more memory and computing power than the **CNN.** Therefore, the **CNN** may be more suitable for lightweight or low-resource applications, while the **ViT** provides better accuracy when sufficient computing resources are available.

7. Conclusion & Future Scope

The **ChiliDoc** project successfully compared a custom **CNN** model and a Vision Transformer **(ViT)** for chili leaf disease classification. The **ViT** achieved a higher accuracy of **99.38%**, while the **CNN** achieved **69.69%**. This shows that the **ViT** can provide better performance for identifying different chili leaf diseases.

Future work will focus on:

- Using lightweight ViT models such as **MobileViT** for mobile and edge devices.
- Increasing the dataset with images captured in different field conditions.
- Developing a mobile application that can provide quick disease classification results to farmers.

8. References (IEEE)

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